KARYOTYPING DIFFERENT ABNORMAL CASES

NUMERICAL ABNORMALITIES

Adan Alnaamneh

CHROMOSOME ABNORMALITIES

Chromosome abnormalities or chromosomal aberration:

Is a disorder characterized by a morphological or numerical alteration in single or multiple chromosomes, affecting autosomes, sex chromosomes, or both.

- Chromosome abnormalities usually involve an error in cell division (mitosis or meiosis), which may occur in the prenatal, postnatal, or preimplantation periods.

- These alterations have significant clinical consequences:

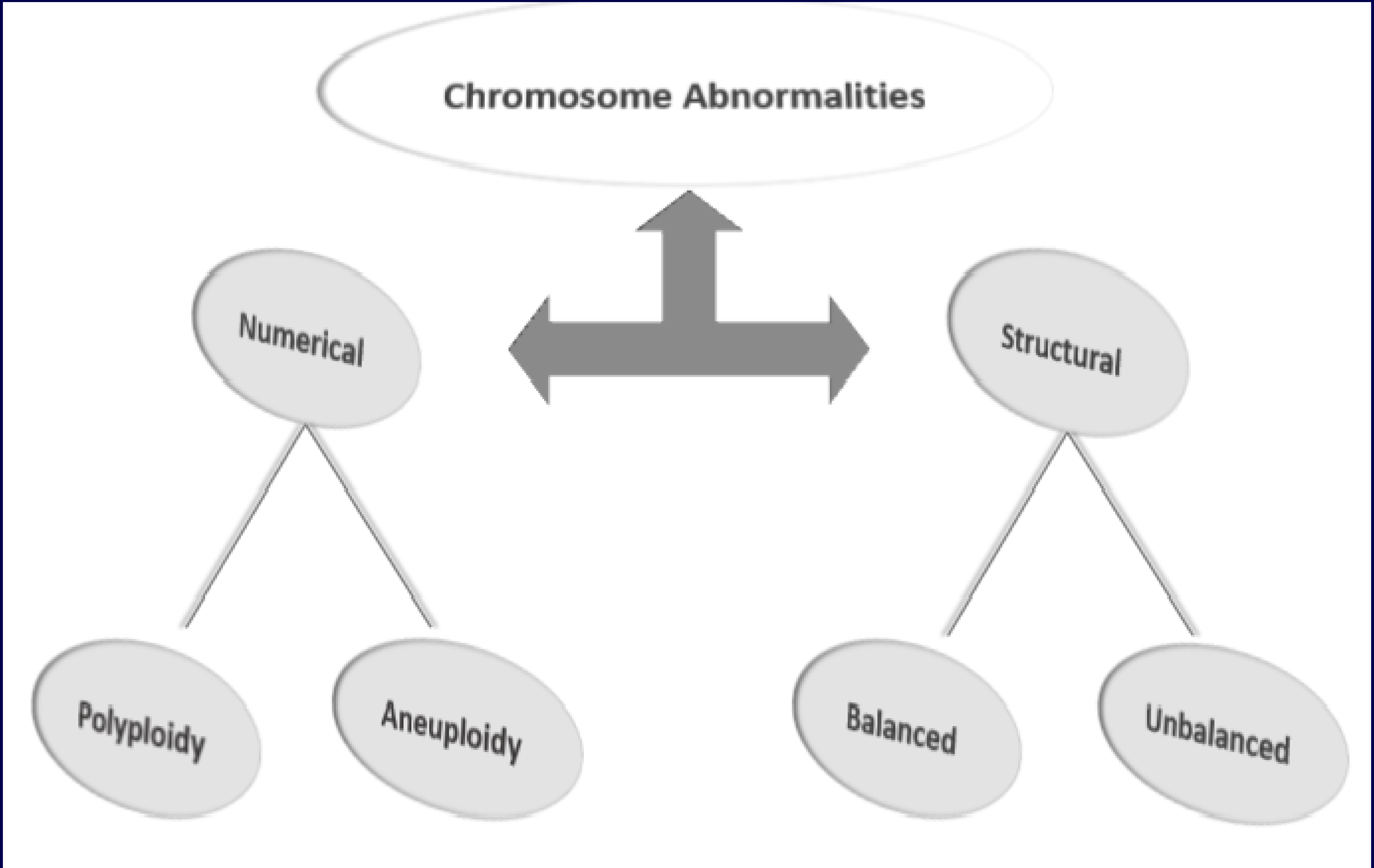
1. Spontaneous abortions
2. Stillbirths
3. Neonatal death
4. Malformations intellectual disability
5. Identifiable syndrome.

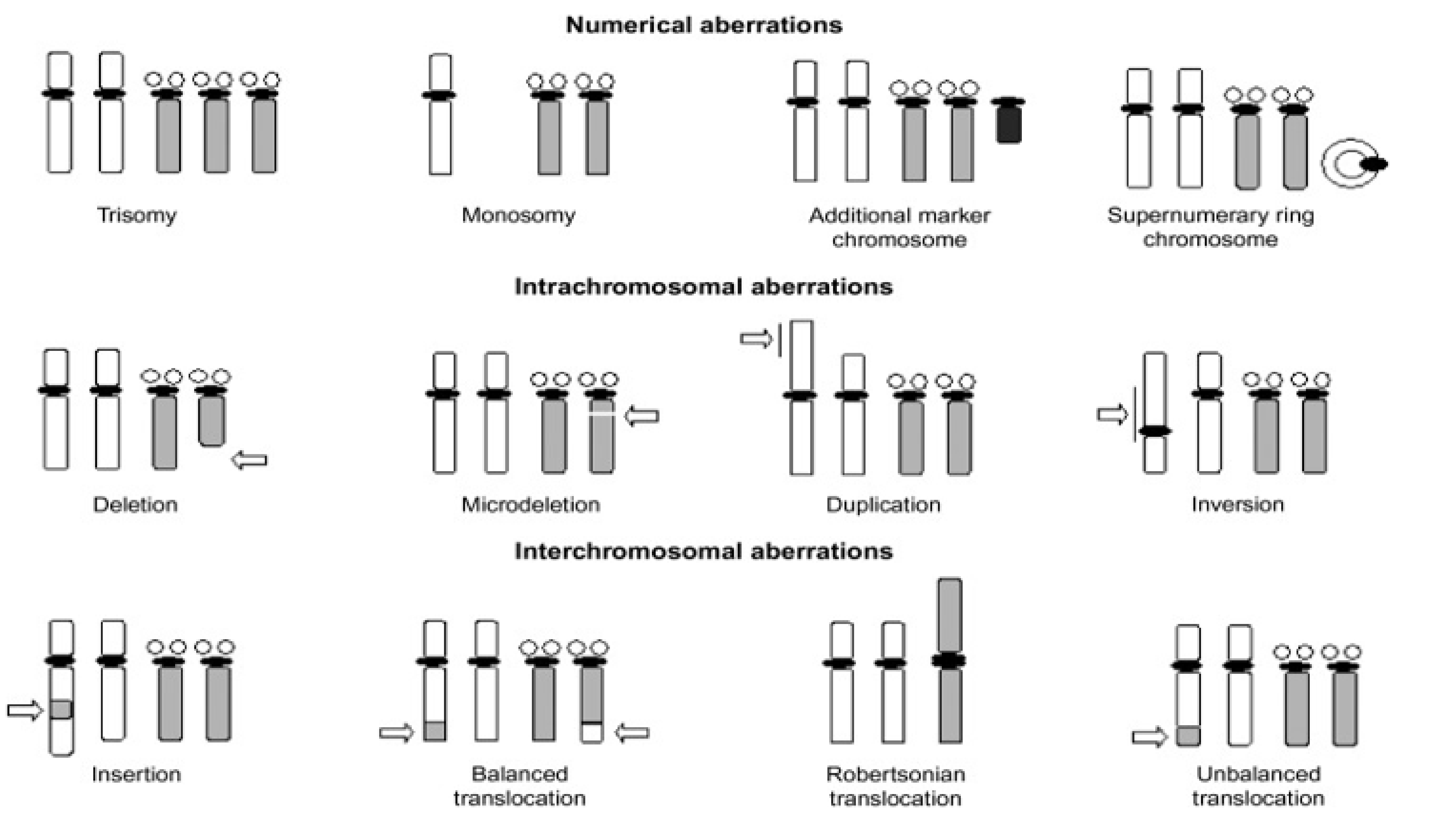
TYPES OF CHROMOSOME ABNORMALITIES

Numerical chromosomal abnormalities: occur when there is a change in the number of chromosomes (more or fewer than 46)

Structural chromosomal abnormalities: result from breakage and incorrect rejoining of chromosomal segments.

Structural can be classified into balanced and unbalanced


Chromosome Abnormalities
Numerical
Structural
Polyploidy
Aneuploidy
Balanced
Unbalanced


Numerical aberrations
Trisomy
Monosomy
Additional marker chromosome
Supernumerary ring chromosome
Intrachromosomal aberrations
Deletion
Microdeletion
Duplication
Inversion
Interchromosomal aberrations
Insertion
Balanced translocation
Robertsonian translocation
Unbalanced translocation

Numerical Chromosomal Abnormalities

Changes in the number of chromosomes:

Polyploidy:

Is the gain of complete extra haploid sets of chromosome.

3n (69 chromosomes), 4n (92 chromosomes), etc.

Aneuploidy:

Is the deviation from the diploid number of chromosomes

2n + 1, 2n -1 etc.

Polyploidy:

A triploid karyogram

Polyploidy:

A tetraploid karyogram

ISCN FOR POLYPLOIDY ABNORMALITIES

Triploid cases:

69, XXX

69, XXY

69, XYY

Tetraploid cases:

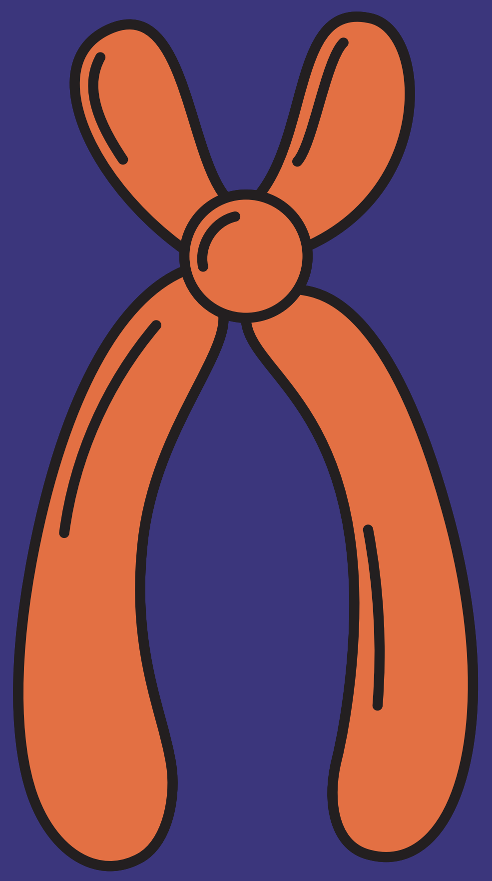
92, XXXX

92, XXYY

Mechanism of Polyploidy

- Failure of pulling apart of 2 chromatids to opposite ends after metaphase stage of mitosis.
- Reduplication of chromosomes without dissolving of nuclear membrane.
- Failure of cytoplasmic division.

ANEUPLOIDY:



- Is the deviation from the diploid number of chromosomes 2n + 1, 2n -1 etc.

- The occurrence of one or more extra or missing chromosomes leading to an unbalanced chromosome complement

Mechanism of Aneuploidy

- Non-disjunction during cell division
- Nondisjunction secondary to chromatid predivision
- Anaphase lag

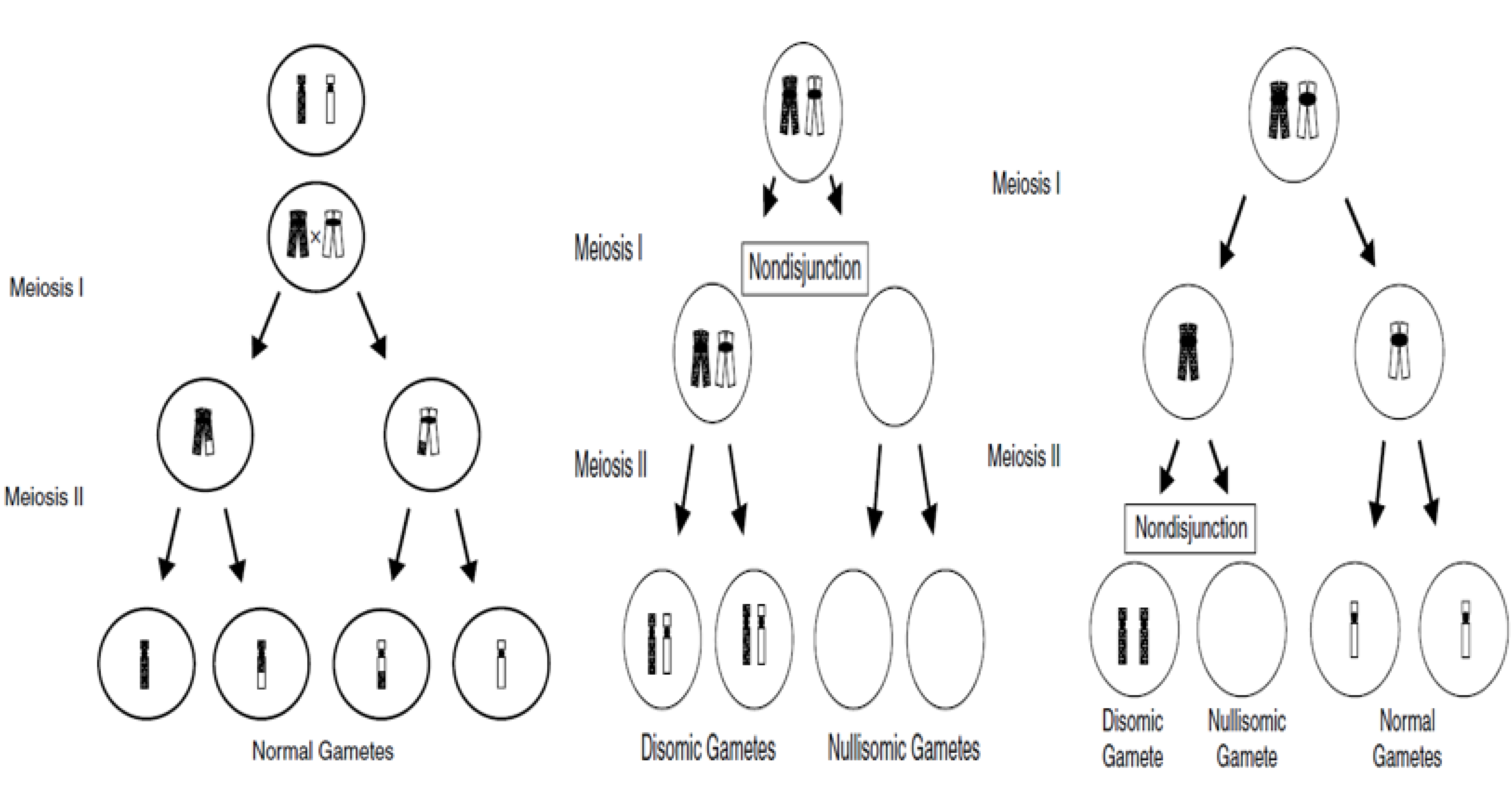
Non-disjunction events

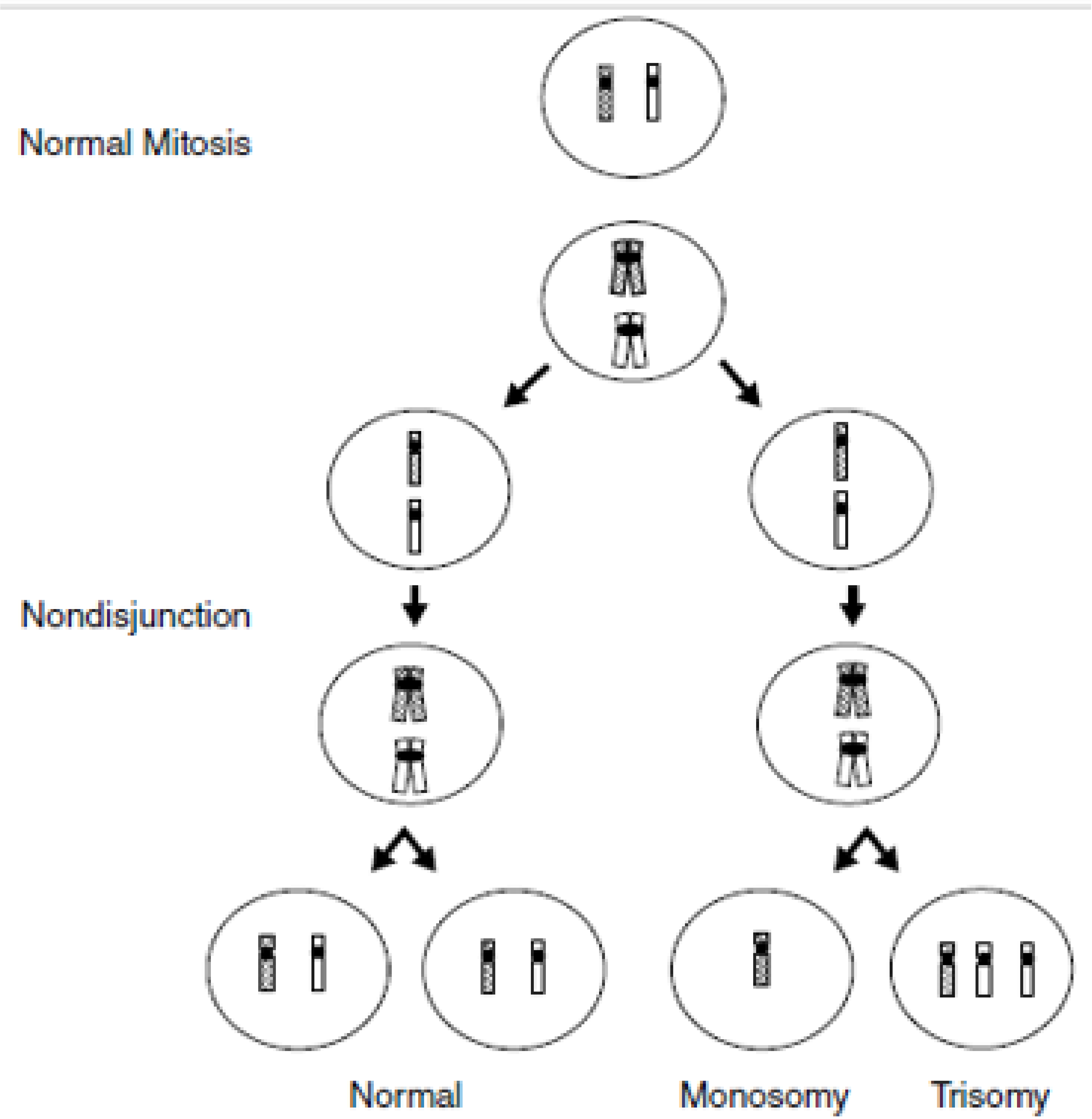
- Non-disjunction: failure of separation of:

1. Homologous chromosomes during meiosis I
2. Sister chromatids during meiosis II

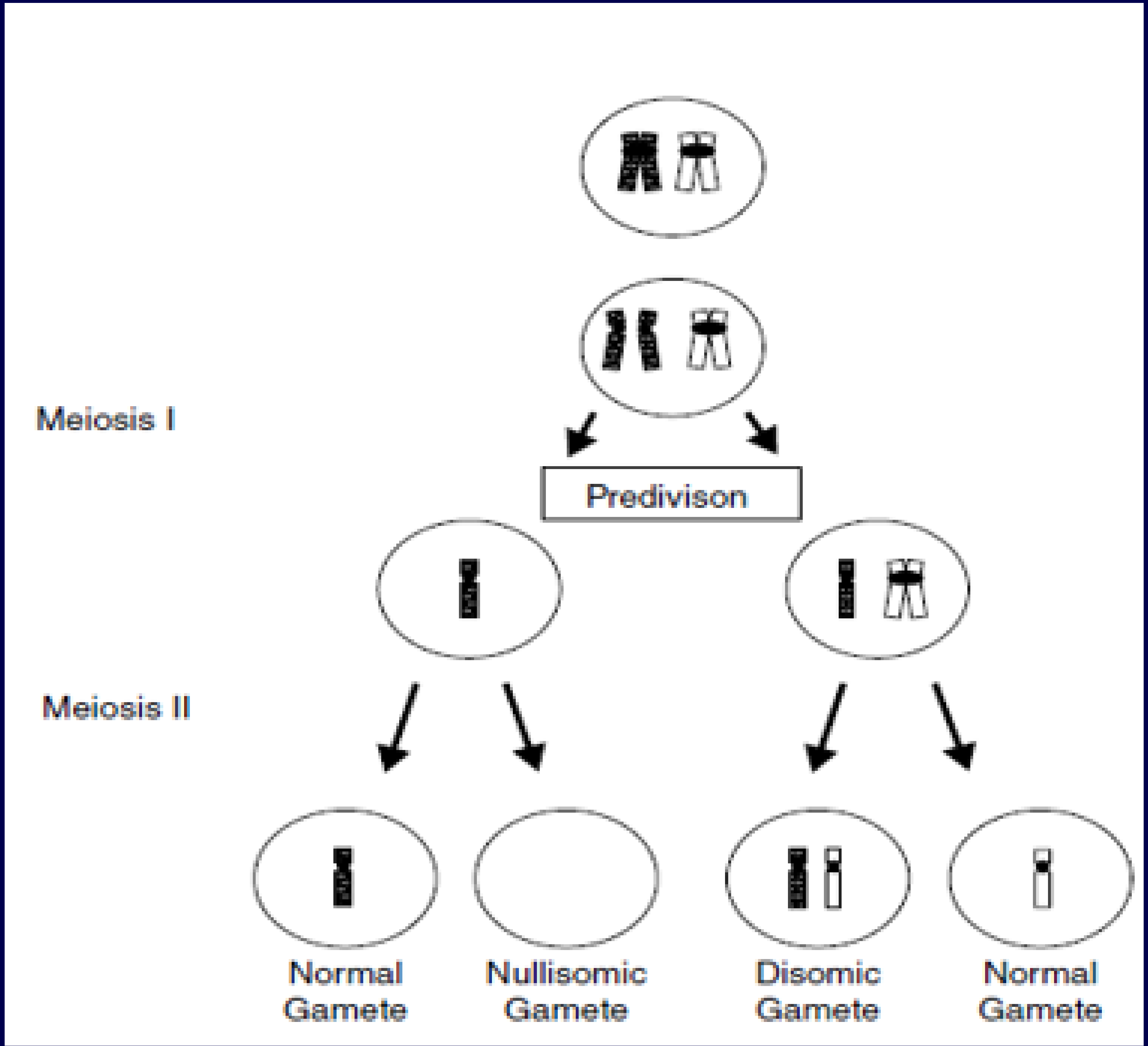
- Fusion of either of these abnormal gametes with a normal gamete can result in trisomy or monosomy

- May involve autosomes or sex chromosomes

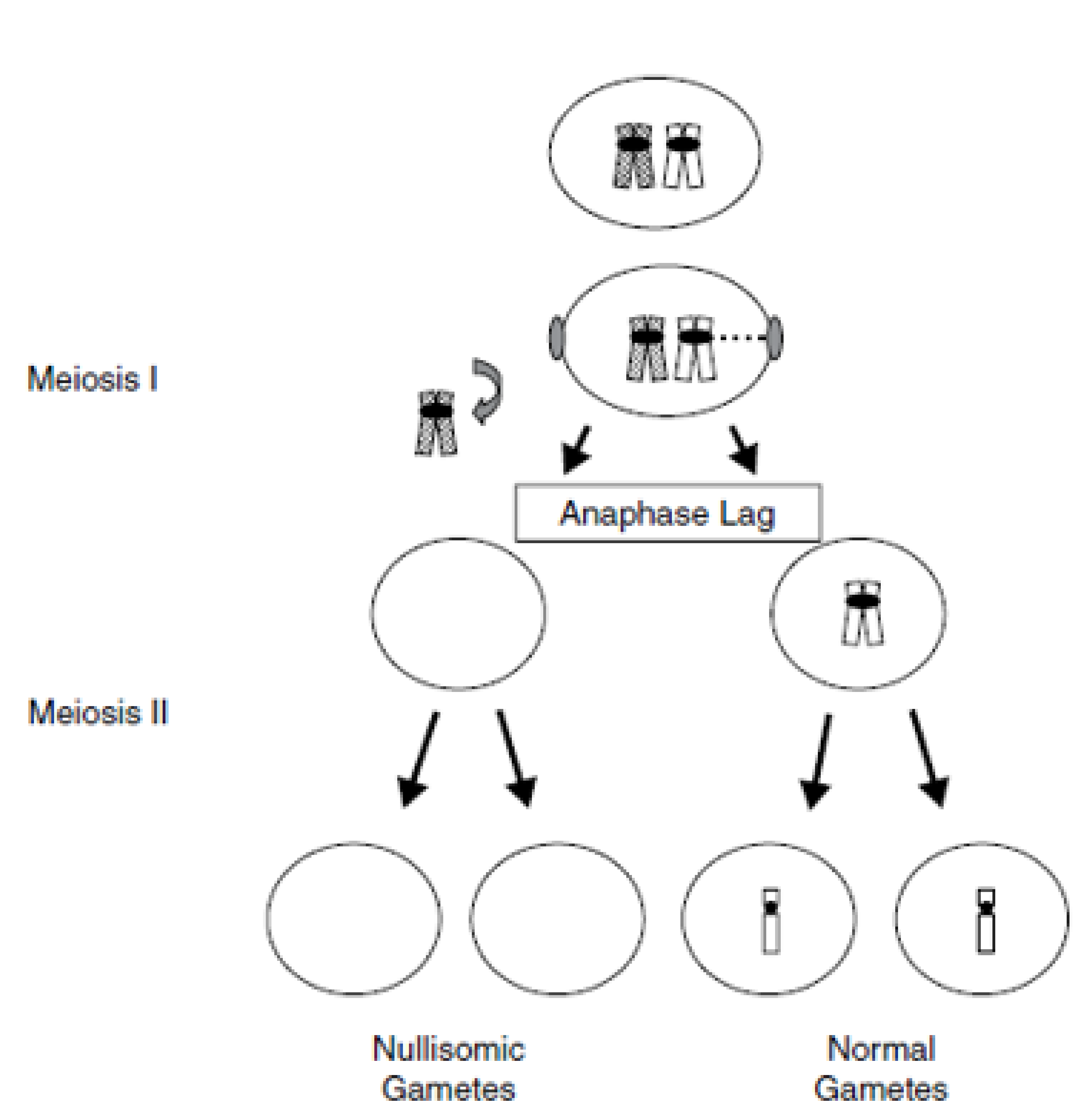

Meiosis I
Meiosis II
Normal Gametes
Meiosis I
Nondisjunction
Meiosis II
Disomic Gametes
Nullisomic Gametes
Meiosis I
Meiosis II
Nondisjunction
Disomic Gamete
Nullisomic Gamete
Normal Gametes


Normal Mitosis
Nondisjunction
Normal
Monosomy
Trisomy

Nondisjunction secondary to chromatid predivision?



Anaphase lag



Trisomies of Chromosomes

Trisomy: Is the Presence of 3 copies of a chromosome

Trisomy of Autosomes: Extra copy of chromosomes (13,18,21)

Trisomy of Sex Chromosomes: (XXX, XXY)

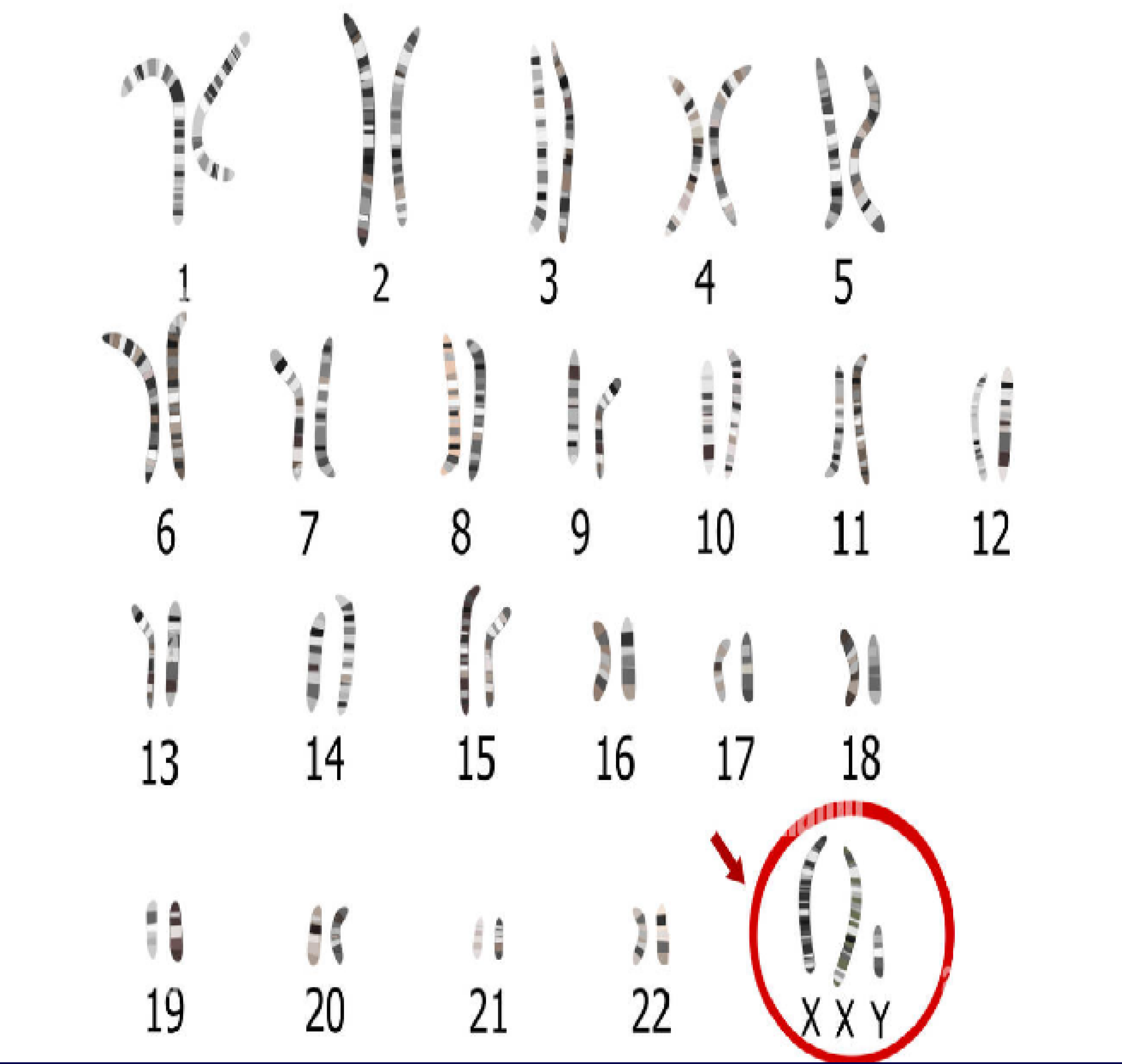
Trisomy of sex chromosomes:

1. Klinefelter Syndrome
2. Meta female

Klinefelter Syndrome

- Male XXY
- Scientific report: 47,XXY
- Main characteristics:

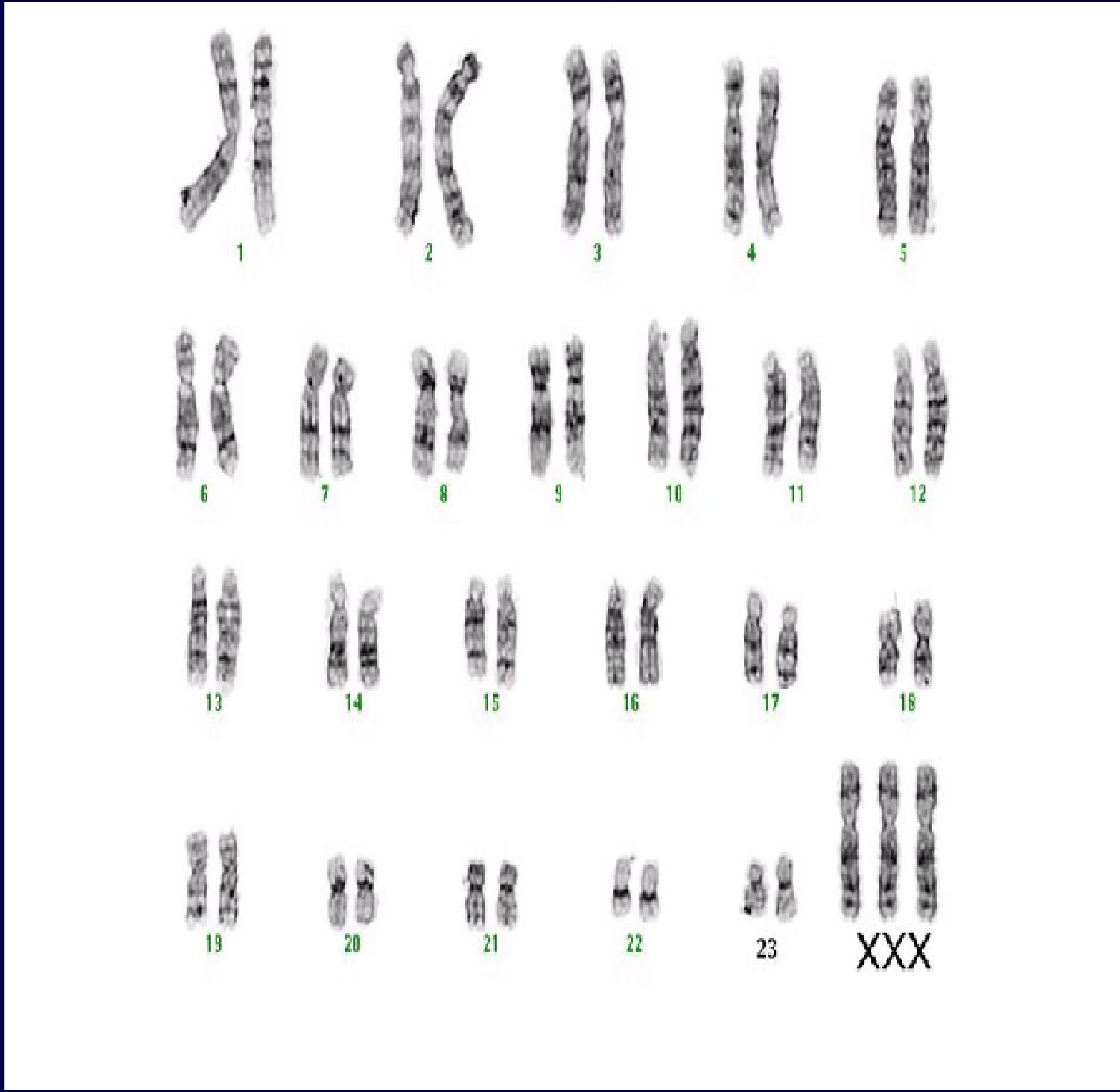
1. Infertile
2. Very tall
3. Large breasts
4. Weak sexual activity
5. Low level of intelligence



Meta female

- Female Triple X (XXX)
- Scientific report: 47,XXX
- Main characteristics:

1. Normal female
2. Small head
3. Tall
4. Speaking difficulties
5. Muscles weakness



Trisomy of Autosomes

- Trisomy 13 (Patau syndrome)
- Trisomy 18 (Edward syndrome)
- Trisomy 21 (Down syndrome)

Patau Syndrome

- Trisomy 13
- Scientific report: 47,XY,+13 or
- 47,XX,+13
- Main characteristics:

1. Mental retardation
2. Heart and eyes Abnormalities
3. small head
4. cleft lip

Edward Syndrome

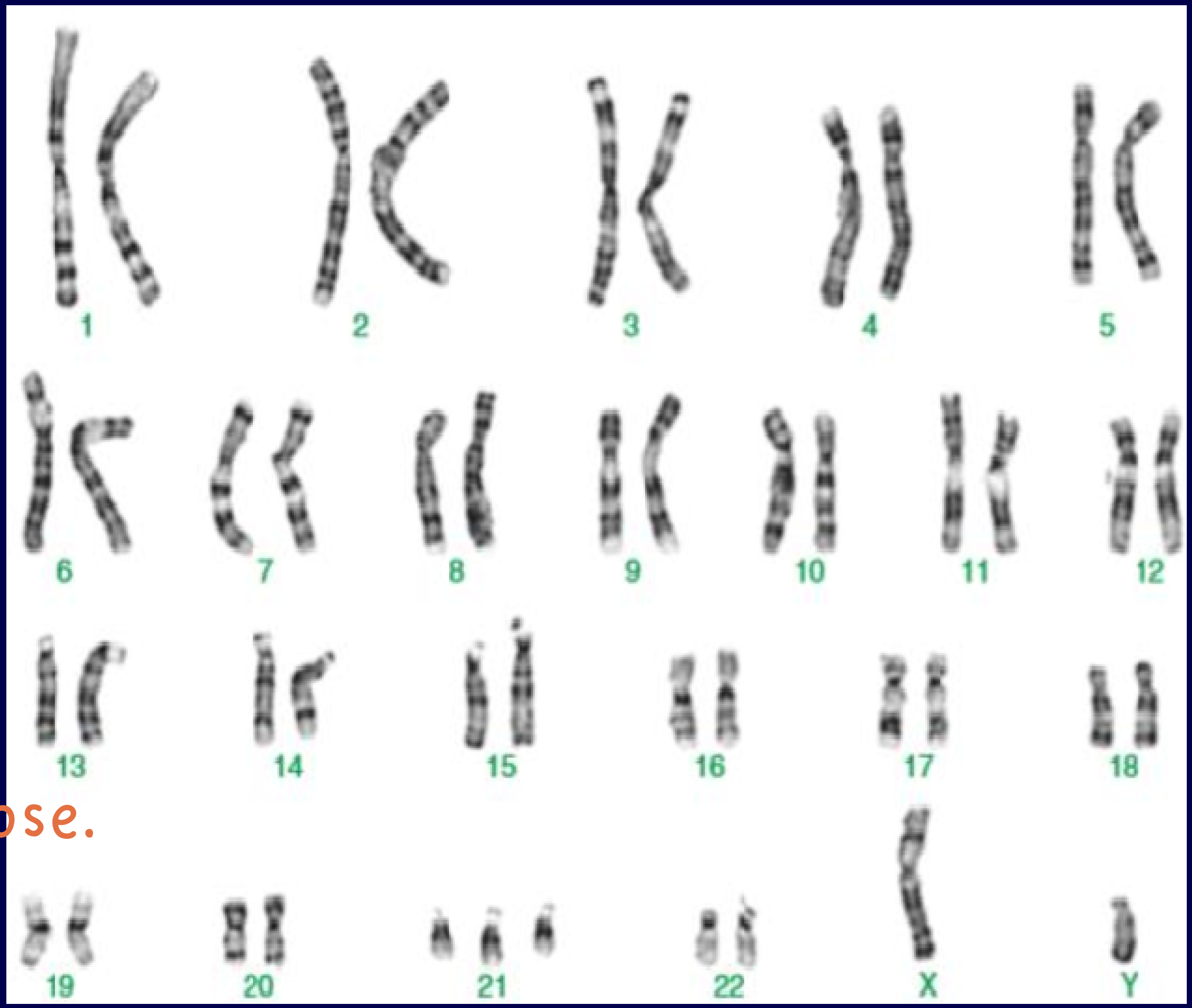
- Trisomy 18
- Scientific report: 47, XX,+18
- or 47,XY,+18
- Main characteristics:

1. Small head
2. Small mouth
3. Abnormal hand
4. Abnormal ears

Down Syndrome

- Trisomy 21
- Scientific report: 47, XX,+21 or
- 47, XY,+21
- Main characteristics:

1. Flattened face, especially the bridge of the nose.
2. Almond-shaped eyes.
3. Short neck and small ears.
4. A tongue that tends to stick out of the mouth.
5. Tiny white spots on the iris.
6. Small hands and feet.



- 48,XX,+13,+21
- 46,XX/47,XX,+21

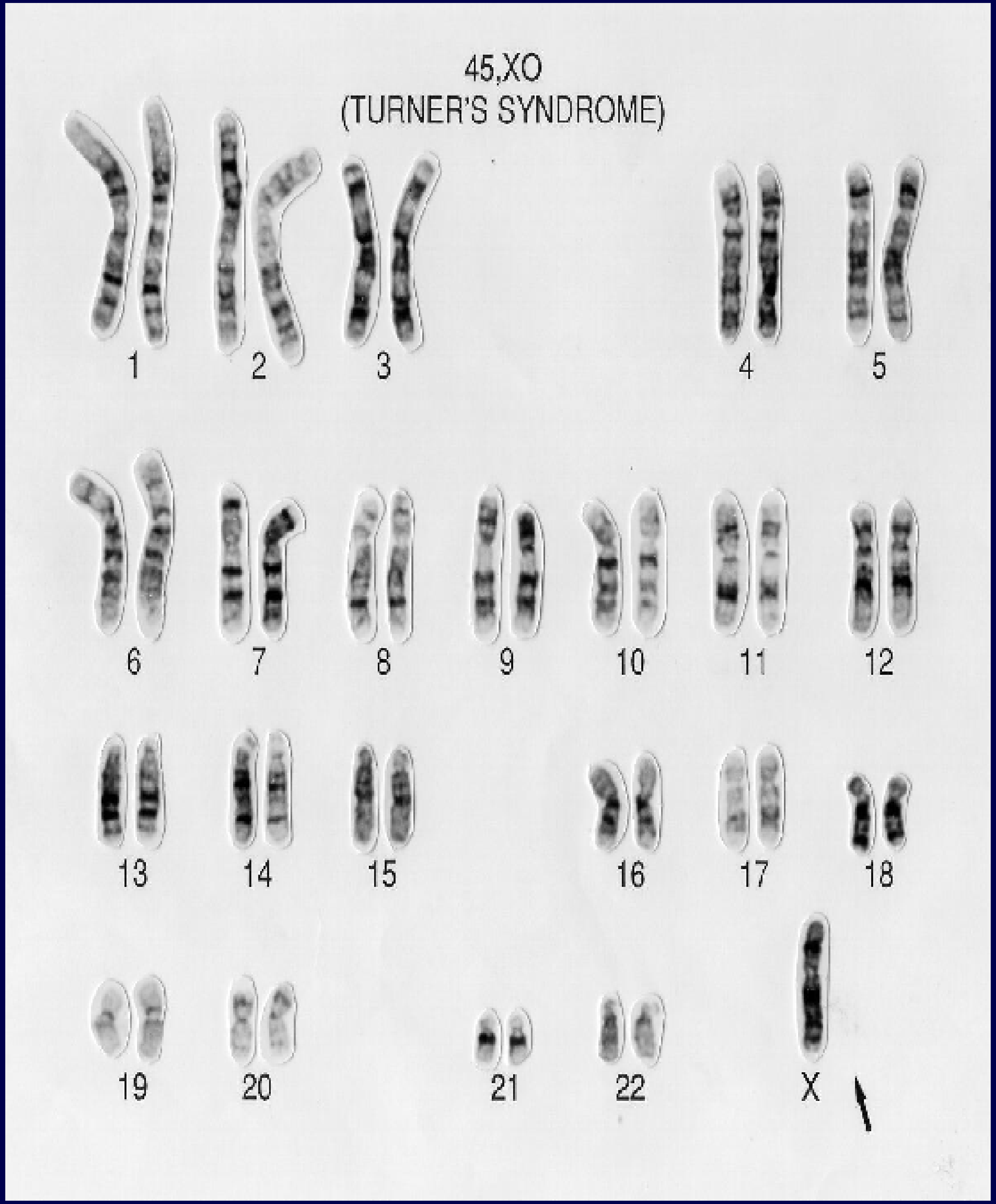
Monosomies of Chromosomes

- Presence of only one member of a chromosome pair in a karyotype
- Can involve autosomes or sex chromosomes
- Usually abort spontaneously
- Monosomy X, is the only survivable and available monosomy.

Turner Syndrome

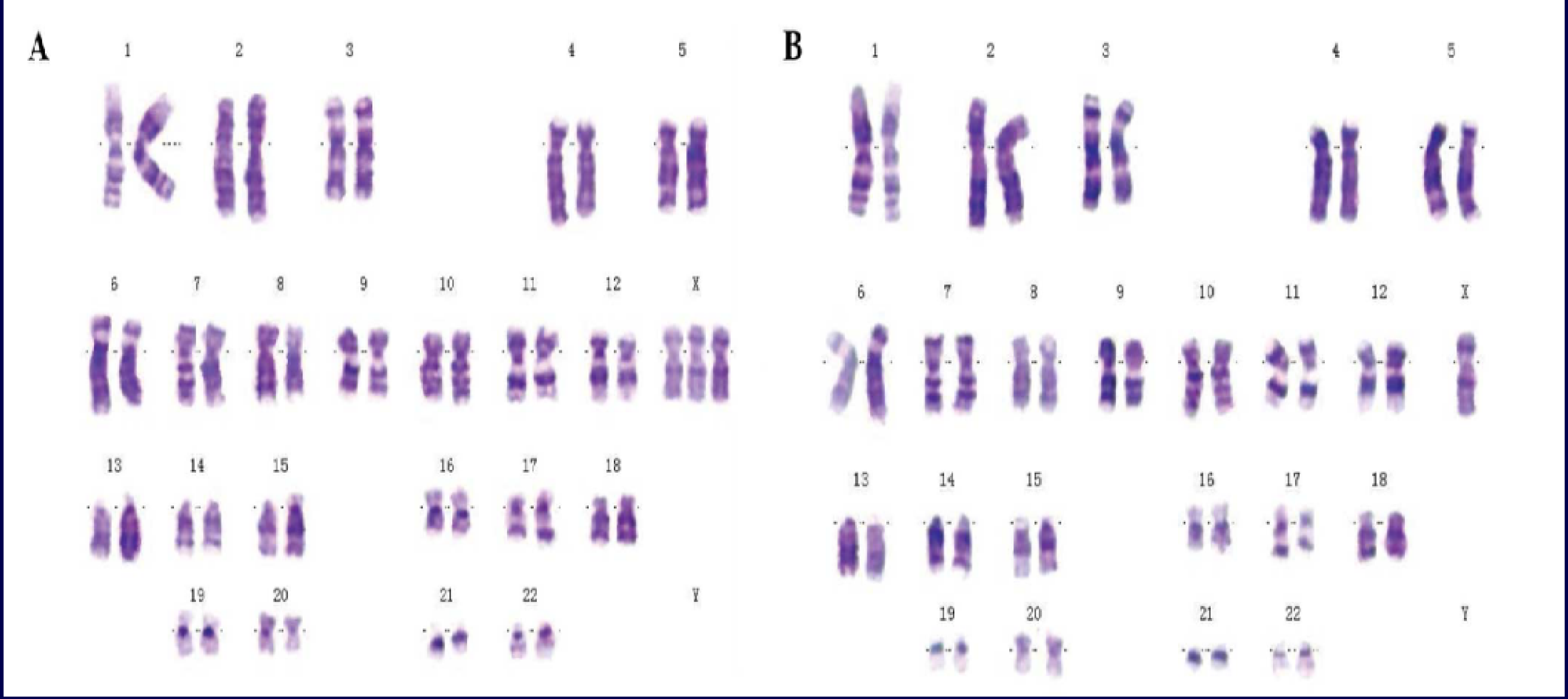
- Monosomy X
- Scientific report: 45, XO
- Main characteristics:

1. Heart and kidney Abnormalities
2. Large legs and hands
3. Short, and small neck
4. infertile



Mosaic turner (some cells turner, some cells normal).

Scientific report: 46, XX / 45, XO



45, XO / 47,XXX

Marker Chromosome

- Marker Chromosome (Undefined chromosome): a structurally abnormal, unidentified extra piece of chromosomal material.
- Sometimes a marker can be characterized with normal karyotyping or FISH.
- Scientific report: 47,XY,+mar